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                        TEST HISTORY & CI
Test Results Tracking, Dashboard, GitHub Actions
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OVERVIEW
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Nucleus maintains a simple test history: an append-only JSON file that records every test run. No hashing, no ledger, no verification – just flat JSON.

History is:

- Automatically created by nucleus test
- Committed to git (or not – your choice)
- Queried with nucleus history command
- Visualized in dashboard as bar chart
- Integrated in CI via GitHub Actions

Flow: nucleus test → tests/test_history.json → nucleus history → Dashboard

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PART 1: TEST HISTORY FILE FORMAT
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```

File: tests/test_history.json (Append-Only)

```
{
  "runs": [
    {
      "timestamp": 1750000000,
      "tests": [
        {
          "name": "led_blink_test",
          "backend": "hardware",
          "status": "pass",
          "detail": "all GPIO toggles verified"
        },
        {
          "name": "led_blink_test",
          "backend": "qemu",
          "status": "pass",
          "detail": "all GPIO toggles verified"
        },
        {
          "name": "uart_loopback",
          "backend": "hardware",
          "status": "skip",
          "detail": "no probe attached"
        }
      ]
    },
    {
      "timestamp": 1750000100,
      "tests": [
        {
          "name": "led_blink_test",
          "backend": "hardware",
          "status": "pass",
          "detail": "all GPIO toggles verified"
        },
        {
          "name": "led_blink_test",
          "backend": "qemu",
          "status": "pass",
          "detail": "all GPIO toggles verified"
        }
      ]
    }
  ]
}
```

Field Meanings:

- timestamp: Unix timestamp (seconds since epoch)
- name: Test name from stm32.toml [[test]].name
- backend: Which backend ran this test ("hardware", "qemu")
- status: "pass" | "fail" | "skip"
- detail: Human-readable reason (failure message, skip reason, etc)

Notes:

- One entry per (test, backend) pair
- Single test run on "both" backends → 2 entries in same run
- Append-only: new runs added, old ones never modified
- Git-friendly: commit it to track history across branches

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PART 2: QUERYING HISTORY
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Command: nucleus history

```
$ nucleus history
```

```
#1  2025-06-15 15:06:40Z  2 passed, 0 failed, 0 skipped
#2  2025-06-15 15:07:15Z  2 passed, 0 failed, 1 skipped
#3  2025-06-15 16:30:22Z  2 passed, 0 failed, 0 skipped
```

Human-readable table of all runs, oldest first. Timestamp, pass/fail/skip counts.

```
Command: nucleus history --last 2
```

```
$ nucleus history --last 2
```

```
#2  2025-06-15 15:07:15Z  2 passed, 0 failed, 1 skipped
#3  2025-06-15 16:30:22Z  2 passed, 0 failed, 0 skipped
```

Last 2 runs only. Useful for CI reports (show recent history, not all 100s).

```
Command: nucleus history --graph
```

```
$ nucleus history --graph
```

```
{
  "schema": "nucleus.history.v1",
  "runs": [
    {
      "timestamp": 1750000000,
      "pass": 2,
      "fail": 0,
      "skip": 0
    },
  ],
}
```

```
jobs:
  nucleus:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Nucleus Check, Build, Test
        uses: harshverma27/nucleus/.github/actions/nucleus@main
        with:
          config: stm32.toml
          build: "true"           # Run nucleus build
          run_tests: "true"      # Run nucleus test (QEMU)
          hardware: "false"      # Skip hardware (no board in CI)
          version: "*"           # Latest nucleus-cli from crates.io
```

```
comment: "true"           # Post summary as PR comment
=====

Action Input Parameters:
=====
| Input      | Default | Purpose                                     |
|-----|-----|-----|
| config     | stm32.toml | Config file to validate                   |
| build      | false    | Run nucleus build                         |
| run_tests  | true     | Run nucleus test (after build)           |
| hardware   | false    | Include hardware test leg (CI: skip)     |
| version    | *        | nucleus-cli version (or "git")          |
| comment    | true     | Post PR comment with summary             |
=====

Outputs:
- conflicts: JSON list of pinout conflicts (from nucleus check)
- firmware-size: Firmware binary size in bytes
- qemu-passed: Number of QEMU tests passed
- qemu-failed: Number of QEMU tests failed
- hardware-passed: Number of hardware tests passed
- hardware-failed: Number of hardware tests failed
=====

Example PR Comment (Auto-Posted):
=====
## ■ Nucleus CI Report

**Validation:** ■ nucleus check passed
**Build:** ■ nucleus build passed (firmware: 47.2 KB)
**Tests (QEMU):** ■ 2 passed, 0 failed

Summary:
- Configuration valid (0 pinout conflicts)
- Firmware compiles successfully
- QEMU tests: 100% pass rate (2/2)
- Hardware tests: skipped (no self-hosted runner)

Test history: [artifact: nucleus-test-history]

[Details...] (shows nucleus history --last 5)

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PART 5: EXAMPLE: COMPLETE CI WORKFLOW
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Scenario: Create PR with new test
=====
1. Local: Write new test in tests/my_test.rs
2. Local: Add [[test]] to stm32.toml
3. Local: nucleus test --backend qemu (passes)
4. Push to GitHub → PR created
5. CI runs (30 seconds):
  - nucleus check (validate config)
  - nucleus build (compile firmware)
  - nucleus test --backend qemu (run tests)
6. CI posts PR comment:
  ■ Validation passed
  ■ Build passed (firmware: 47.2 KB)
  ■ Tests: 5 passed, 0 failed
7. Merge approved
8. Local: nucleus test --backend hardware (validate on real board)
9. Manual: nucleus flash (deploy to production)
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PART 6: PERSISTING HISTORY
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Commit History File (Optional):
=====
$ git add tests/test_history.json
$ git commit -m "test results: 2 passed, 0 failed"

Benefits:
- Track test history across branches
- Revert to old history if needed
- CI can compare against parent commit
- Blame/log shows when tests started failing

Gitignore Pattern (Alternative):
=====
# .gitignore
tests/test_history.json      # Don't commit history (ephemeral)

Benefits:
- History is local-only, per developer
- No merge conflicts
- CI creates fresh history per run

Recommendation: Commit it. History is useful context, not noise.
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PART 7: TROUBLESHOOTING
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No test_history.json file:
  Run: nucleus test (at least once)
  File is created on first run.

History shows all skipped:
  Check: Are tests configured with correct backend?
  - backend = "qemu" runs on QEMU only
  - backend = "hardware" requires real board
  Use: backend = "both" (default) for maximum coverage

CI not posting comments:
  Check: GitHub token permissions (read:contents, write:pull-requests)
  Check: comment: "true" in action inputs
  Fallback: Job Summary visible even without comment permissions

Tests pass locally, fail in CI:
  Likely: Timing, environment differences
  Fix: Add delays, check for QEMU-specific behavior
  Debug: nucleus history --graph (compare local vs CI artifact)
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PART 8: COMMANDS REFERENCE
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View history:
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```
nucleus history --last 5           # All runs (human table)
nucleus history --last 5 runs      # Last 5 runs
nucleus history --graph            # JSON for dashboard/export

Show details:
nucleus show                       # Latest run details
nucleus show 3                     # Run #3 details

Manage history:
rm tests/test_history.json         # Start fresh
git log tests/test_history.json    # History of history changes

Create test:
# 1. Add [[test]] to stm32.toml
# 2. Write #[test] function
nucleus test --list                # See all tests
nucleus test my_test               # Run specific test
nucleus test --backend hardware   # Hardware only

CI:
# GitHub Actions auto-runs on push/PR
# Output: PR comment + artifact: nucleus-test-history
# Control via .github/workflows/ci.yml
```